

This will take some time. If the process is successful, you will have a new command prompt, and you can now eject the SD card.

Booting PandaBoard

Install *minicom* or *Gtkterm* (I personally prefer *minicom*) on the desktop—in the terminal, run `sudo apt-get install minicom`. Launch it—for example, `sudo minicom -s`. To configure it, select the correct USB port, set the baud rate to 115200, stop bits to 1 and parity bits, software control and hardware control to 'NONE'.

Now insert the SD card into PandaBoard.

Connect the HDMI/DVI cable to the monitor and PandaBoard. Attach the USB hub to PandaBoard, and connect your mouse and keyboard to the hub. Connect the serial cable to your laptop or desktop. Now power up PandaBoard. If all went well, you should see a familiar Ubuntu installation screen.

Installing Ubuntu 12.04

The installation procedure is quite simple—just follow the on-screen instructions. Mind you, this process will take a very long time, so be patient.

Speeding up PandaBoard

There are a few tweaks you can do to increase the performance of Ubuntu on PandaBoard. First, disable swap, which is nothing but virtual RAM. So, edit `/etc/fstab` and put a '#' in front of the line that creates a swap file. Now, for transient data, use the *tmpfs* filesystem, which uses RAM, and hence provides much faster response compared to the SD card. Edit `/etc/fstab` and add the following lines:

```
tmpfs /tmp                tmpfs nodev,nosuid      0 0
tmpfs /var/log            tmpfs nodev,nosuid      0 0
```



Figure 1: The board



Figure 2: My hack area

Installing Xubuntu

You'll find Ubuntu 12.04 on PandaBoard quite slow. We need a lightweight desktop environment. Now Xubuntu is a derivative of Ubuntu, but with Xfce as the desktop environment plus some other applications. Let's install Xubuntu on top of Ubuntu. In a terminal, run the following:

```
sudo apt-get update
sudo apt-get install xubuntu-desktop
```

Now, this process will take very long. Once it's done, restart the system and while logging in, select the 'Xubuntu Desktop' to see the new desktop. You can also update your Xubuntu desktop; in a while you will have a faster-running system on the small but powerful PandaBoard.

If you'd like to see this hacker's workspace, take a look at Figure 2, which shows my set-up, with PandaBoard atop the laptop cooling pad and connected as mentioned above.

In the forthcoming articles in this series, I will discuss how to port other OSs to PandaBoard, how to do image processing on PandaBoard, and various other interesting topics.

Troubleshooting

It is quite possible that you will face some errors while installing Ubuntu or Xubuntu on PandaBoard. In case you face an error you cannot solve, post your problem on the PandaBoard Google Group (<http://groups.google.com/group/pandaboard>). The community behind this group is very nice and friendly and will help you out—they also reply very quickly. 

Reference links

- [1] http://omappedia.org/wiki/PandaBoard_Ubuntu_How-tos/
- [2] <http://pandaboard.org/>
- [3] <https://wiki.ubuntu.com/ARM/OmapDesktopInstall>
- [4] <http://www.svtronics.com/ecommerce/panda>

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